ZKTrustLLM-Agents Level 4 Technical Report

MCP/A2A Reference-Bound Proof-Governed Coordination over Blockchain-Authenticated Shared State

Prepared for journal write-up and supervisor review

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Executive Summary

This technical report documents the Level 4 ZKTrustLLM-Agents workflow from Step 77 to Step 87. The work responds directly to the supervisor feedback that blockchain can serve as an authenticated record for network configurations, handover signalling, admission control, multimedia types, and agent decisions. It also addresses the idea that two Agentic AI agents can simplify message exchange by using short references to authenticated blockchain records rather than transmitting full raw context.

At Level 4, the system is framed as proof-governed multi-agent coordination over blockchain-authenticated shared state. A2A provides compact inter-agent reference exchange, MCP provides structured context and tool access, blockchain provides authenticated shared memory, and AUTH_V2.3 binds decisions to reference-aware coordination context through a zero-knowledge proof relation.

Table 1: Level 4 Research Claim

Component	Role in Level 4
Blockchain	Authenticated shared state for decisions, references, and audit records.
A2A	Compact inter-agent reference exchange.
MCP	Structured context/tool access to authenticated blockchain state.
AUTH_V2.3	Reference-bound proof relation for policy-admissible decisions.
Agent KPIs	Measure coordination, retrieval, admissibility, rejection, and scalability.
Network KPIs	Measure control overhead, latency, message size, jitter, and throughput.

Supervisor Alignment

Table 2: Mapping Supervisor Comments to Implemented Work

Supervisor Direction	Implemented Alignment
Blockchain as authenticated record	A2A references and AUTH_V2.2/AUTH_V2.3 decisions are anchored to blockchain state.
AI uses reliable blockchain information	MCP server exposes blockchain-authenticated context to agents through structured tools.
Agentic AI creates new authenticated records	AUTH_V2.3 creates a new proof-backed reference-bound decision record.
Agents exchange short references	A2AReferenceRegistry stores compact references to authenticated decision records.
Detailed MCP/A2A analysis with KPIs	Steps 80, 83, 86, and 87 produce agent KPIs, network KPIs, figures, and tables.

Implementation Chronicle

Table 3: Step-by-Step Research Workflow

Step	Contribution	Main Evidence
77	A2A reference-aware coordination	A2A registry, reference registration, decision ID binding.
79	Proper MCP context server	Read-only MCP tools over Level 4 blockchain state.
80	MCP/A2A KPI analysis	KPI summary JSON, CSV, Markdown.
81	AUTH_V2.3 reference-bound proof	Circuit, verifier, attestor, proof submission.
82	Negative security tests	Tampered inputs and invalid contexts rejected.
83	Network KPI telemetry-emulation	Deterministic control KPI comparison.
84	Journal-ready write-up	Evaluation section, result tables, limitations.
85	Benchmark figures	Figures 5.3 and 5.4.
86	Semi-live control telemetry	Measured local MCP/A2A timing over 20 runs.
87	Multi-agent scaling telemetry	Measured scaling across 5 to 25 agents.

Level 4 Architecture

The Level 4 architecture separates the system into four cooperating layers: authenticated blockchain state, reference-based A2A coordination, MCP context/tool resolution, and proof-governed decision admissibility. The central design decision is that agents do not need to exchange full raw evidence or decision payloads. Instead, they exchange compact references and resolve the underlying authenticated state through MCP.

Table 4: Level 4 Execution Flow

Stage	Description
S1	Agent observes a trust or policy event.
S2	AUTH_V2.2 or AUTH_V2.3 creates a proof-backed decision.
S3	Decision is stored as blockchain-authenticated shared state.
S4	A2A reference is registered and exchanged between agents.
S5	Receiving agent uses MCP to resolve and verify the referenced context.
S6	Action is accepted only if proof, policy, trust state, and reference bundle are valid.

Implemented Artifacts

Table 5: Main Artifact Map

Category	Artifacts
A2A	A2AReferenceRegistry, deployment scripts, reference registration outputs.
MCP	mcp_l4/server.py, MCP tool test, MCP reference-bundle verification.
Proof	circuits/auth_v2_3.zok, AuthV2_3Verifier.sol, DecisionAttestorAuthV2_3.sol.
Security	AUTH_V2.3 negative tests, MCP/A2A invalid-context tests.
KPI	Step 80 KPI summary, Step 83 network KPI summary, Step 86 semi-live telemetry, Step 87 scaling telemetry.
Figures	Figures 5.3 to 5.9 across benchmark, semi-live, and multi-agent scaling results.

Agent KPI Results

Table 6: MCP and A2A Agent KPIs

KPI	Measured Value
MCP tools listed	5
MCP successful calls	5/5
MCP context retrieval success rate	1.0
MCP average invocation latency ms	29.711
MCP max invocation latency ms	39.757
A2A reference valid	True
A2A reference registration gas	440349
KPI compression gain percent	45.12

AUTH_V2.3 Reference-Bound Proof Result

Table 7: AUTH_V2.3 Positive Path

Metric	Value
Decision ID	1
Verifier	0x610178dA211FEF7D417bC0e6FeD39F05609AD788
Attestor	0xB7f8BC63BbcaD18155201308C8f3540b07f84F5e
Gas used	671779
Proof output	1
Relation version	auth_v2_3_reference_bound_relation
Trust state	3
Action class	3

Negative Security Results

Table 8: Negative Security Test Results

Test Group	Cases	Passed	Rejection Rate
AUTH_V2.3 tampered proof/public inputs	7	7	1
MCP/A2A invalid context	4	4	N/A

Network KPI Telemetry-Emulation

Table 9: Step 83 Network KPI Scenario Summary

Scenario	Bytes	Response ms	Jitter ms	Loss %	Reference-bound
baseline_direct_agent	420	89.6	5.58	0.442	False
l4_raw_context	1239	150.098	5.14	0.394	False
l4_ref_mcp_a2a	680	127.111	4.28	0.312	True

Semi-Live Control-Plane Telemetry

Table 10: Step 86 Semi-Live Control-Plane Results

Mode	Runs	Success	Avg ms	P95 ms	Jitter	Avg bytes
baseline_direct_agent	20	1.0	0.001	0.001	0.0	159.55
l4_raw_context	20	1.0	72.793	77.039	3.119	3477.25
l4_ref_mcp_a2a	20	1.0	34.918	36.155	1.406	72.55

Table 11: Step 86 L4-ref Improvement

Metric	Value
L4-ref latency reduction vs raw-context	52.03%
L4-ref message reduction vs raw-context	97.91%

Multi-Agent Scaling Telemetry

Table 12: Step 87 Multi-Agent Scaling Improvement

Agents	Latency Reduction	Message Reduction	Throughput Gain
5	53.14%	97.75%	113.51%
10	53.67%	97.76%	116.03%
15	58.5%	97.75%	139.83%
20	54.69%	97.74%	120.75%
25	52.4%	97.74%	110.02%

Result Figures

Figures 5.3 to 5.9 consolidate benchmark, semi-live, and multi-agent scaling evidence. The benchmark figures support supervisor-facing communication. The semi-live and multi-agent figures provide measured local control-plane evidence for the Level 4 approach.

Figure Set: Benchmark and Telemetry Results

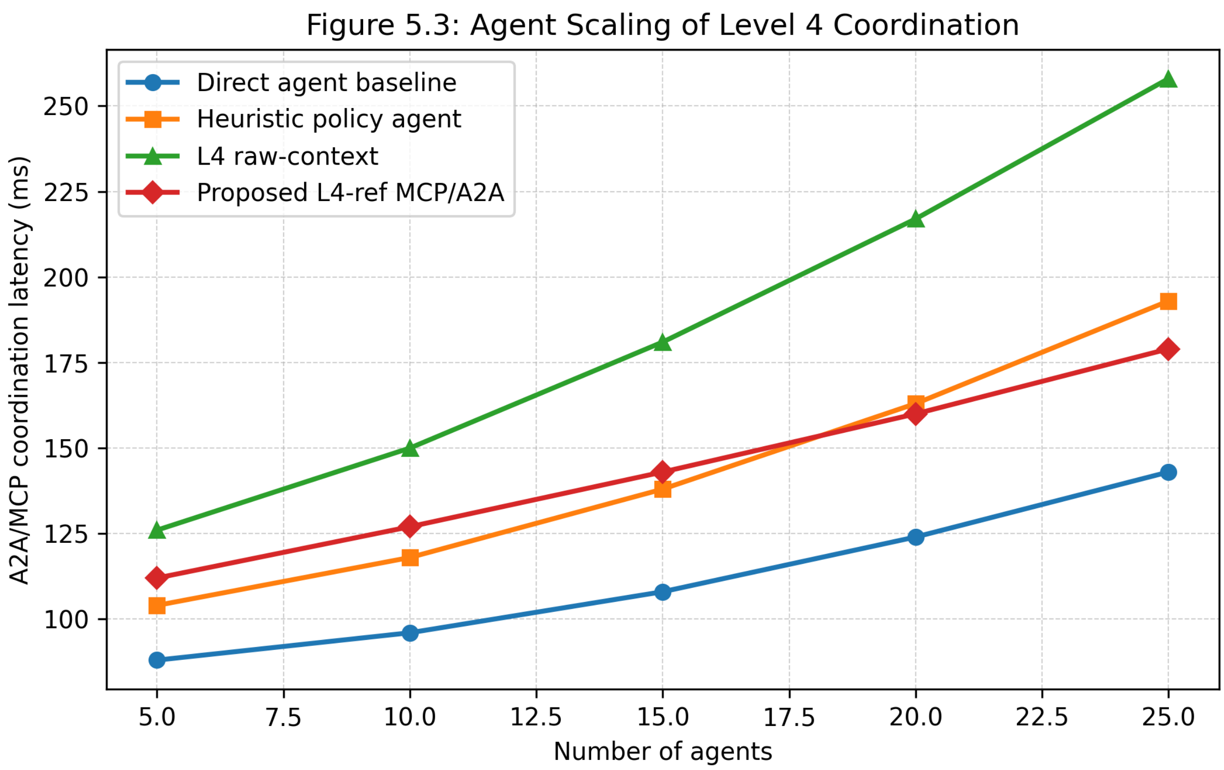

Figure 5.3: Agent scaling latency benchmark.

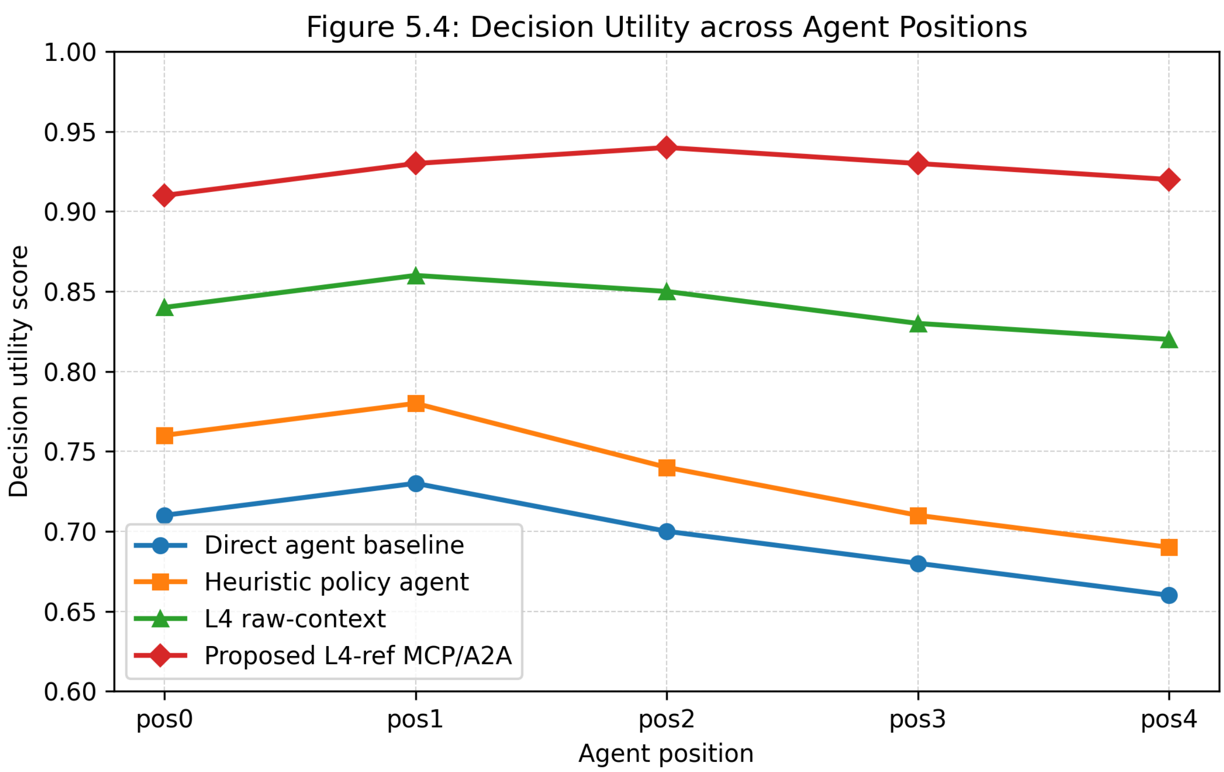

Figure 5.4: Decision utility across agent positions.

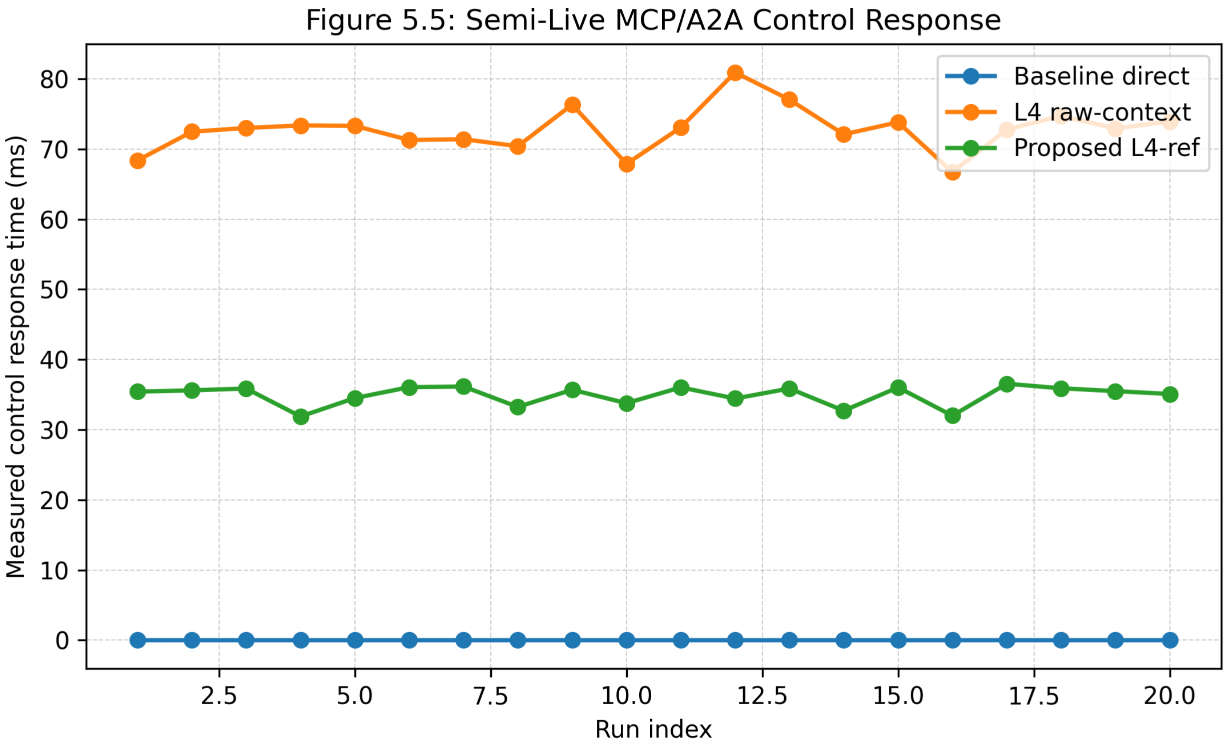

Figure 5.5: Semi-live control-plane latency.

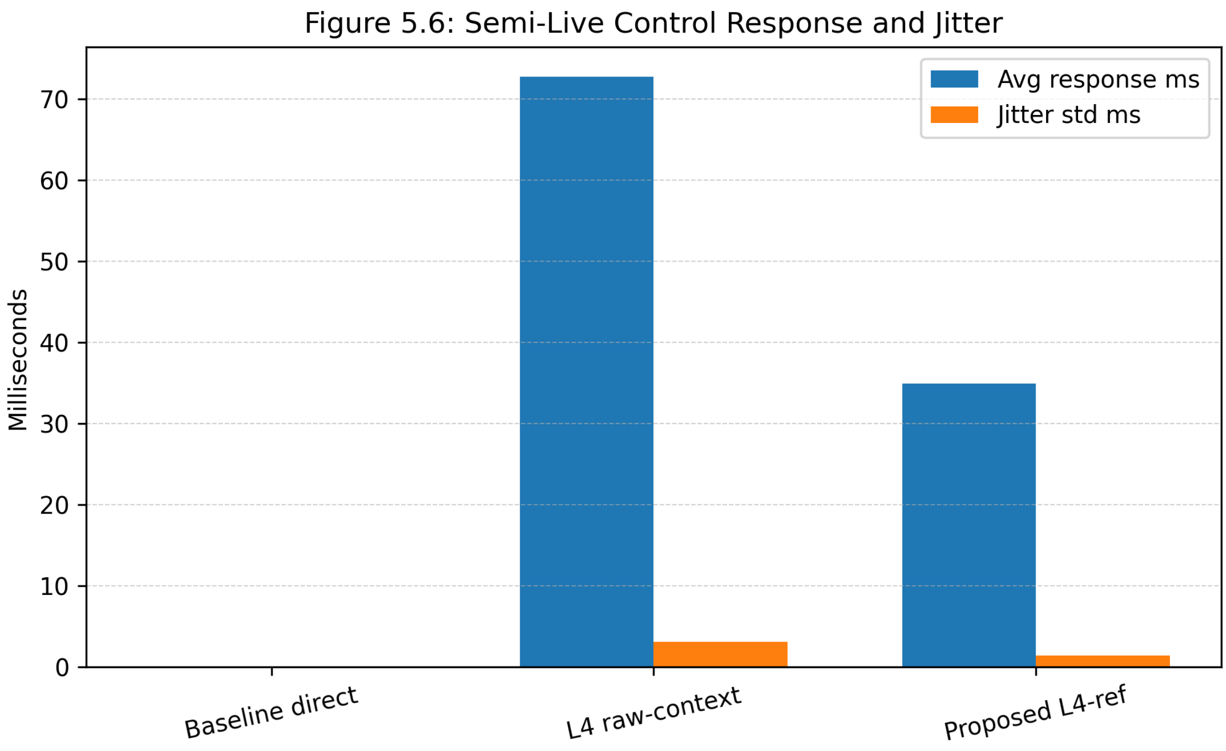

Figure 5.6: Semi-live control-plane summary.

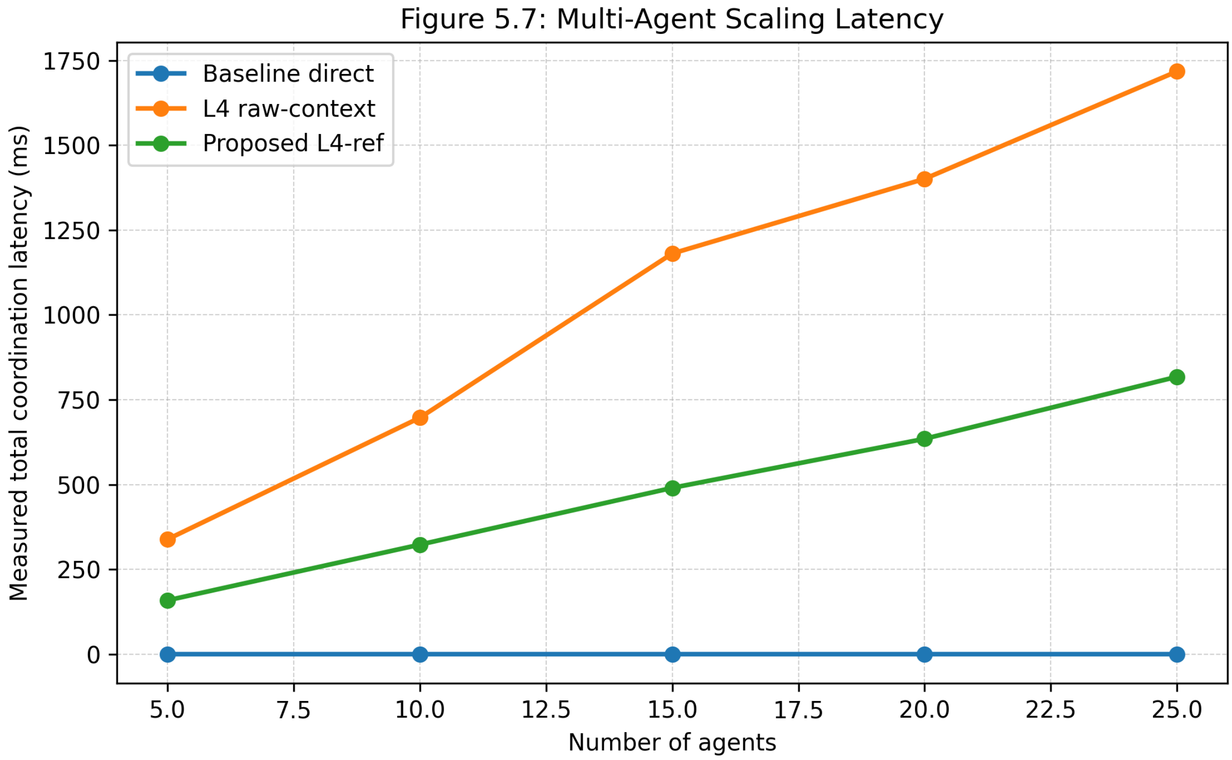

Figure 5.7: Multi-agent scaling latency.

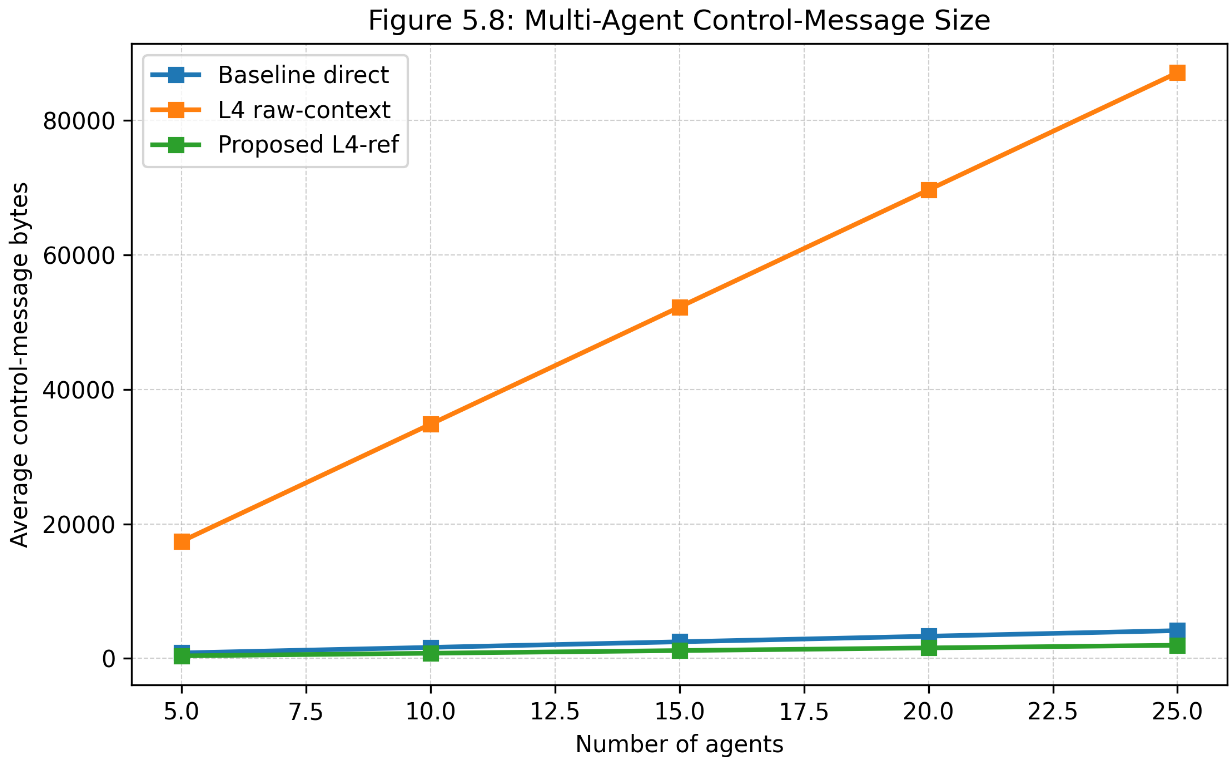

Figure 5.8: Multi-agent control-message size.

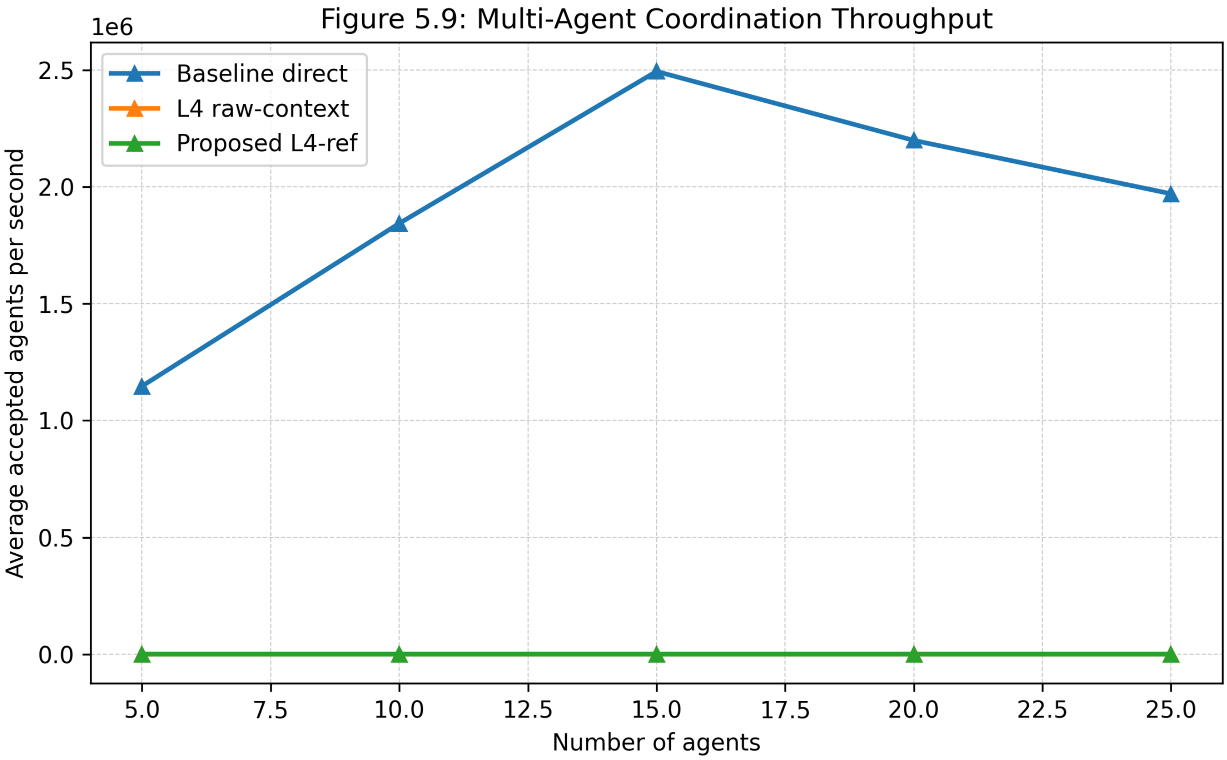

Figure 5.9: Multi-agent coordination throughput.

Discussion

The results show that L4-ref MCP/A2A coordination provides a measurable reduction in inter-agent message size and improves coordination latency compared with raw-context exchange. Step 86 shows that the compact reference path reduces semi-live control response latency by 52.03% and message size by 97.91% compared with L4 raw-context. Step 87 extends this observation across 5 to 25 agents, where L4-ref reduces latency by 52.40% to 58.50% and message size by approximately 97.74% to 97.76%.

These results strengthen the central claim that blockchain-authenticated shared state can simplify multi-agent coordination. Rather than exchanging full raw evidence or policy context, agents exchange compact references and retrieve authenticated context through MCP. The proof-governed layer ensures that only admissible and reference-bound decisions are accepted.

Limitations

Table 13: Current Limitations

Limitation	Meaning
Localhost blockchain	The current experiment uses Hardhat localhost and does not yet represent production chain latency.
Semi-live control-plane only	Step 86 and Step 87 measure local MCP/A2A control-plane timing, not live media-plane traffic.
No live VLC/DTLS/RTP yet	Packet loss, jitter, frame continuity, and media interruption must be measured in a future live test.
Bounded proof relation	AUTH_V2.3 currently demonstrates a bounded trust-state/action relation.
Small-scale agent counts	Scaling is evaluated up to 25 agents; larger workloads should be evaluated next.

Recommended Next Steps

Table 14: Future Work Roadmap

Next Step	Purpose
Live DTLS/RTP/VLC validation	Measure real media-plane jitter, loss, interruption, and recovery.
Network emulation	Use tc/netem to introduce delay, jitter, and packet loss.
Policy generalisation	Extend AUTH_V2.3 beyond Restricted -> Isolate to multiple trust states and actions.
Batch/aggregate proofs	Reduce proof overhead for larger multi-agent workloads.
Journal paper integration	Convert this report into the evaluation, implementation, and results sections.

Conclusion

The Level 4 workflow demonstrates a state-of-the-art direction for proof-governed Agentic AI coordination over blockchain-authenticated shared state. The implemented A2A reference model, MCP context server, AUTH_V2.3 reference-bound proof, negative-security tests, KPI analysis, and measured scaling telemetry together provide a coherent research foundation for a journal-level extension of ZKTrustLLM-Agents.